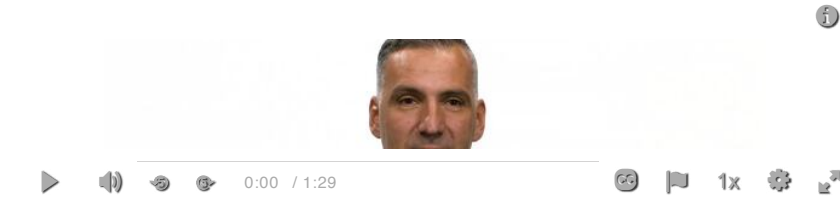


Lesson Six Overview



Throughout the lecture, we are giving you some **“food for thought”** -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- Understand the problem of finding individually important nodes/edges or groups of important nodes/edges
- Define and analyze common centrality metrics
- Define and analyze common “network core” metrics
- Learn how to apply centrality metrics and core identification algorithms through case studies

Required and Recommended Reading

Required Reading

- Chapter 7 (mostly sections 7.1-7.8) - M.E.J.Newman, [Networks: An Introduction](https://www.amazon.com/Networks-Mark-Newman/dp/0198805098) , Oxford University press.
- Section 14.3 - D. Easley and J. Kleinberg, [Networks, Crowds and Markets](https://www.cambridge.org/core/books/networks-crowds-and-markets/A70C7855A3003FE1079C25F8397AF641) , Cambridge Univ Press, 2010 (also available [online](https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book.pdf) , <https://www.cs.cornell.edu/home/kleinber/networks-book/networks-book.pdf>.)
- **“Rich-clubness test: how to determine whether a complex network has or doesn’t have a rich-club?”**  <https://arxiv.org/abs/1704.03526> By Alessandro Muscoloni and Carlo Vittorio Cannistraci

Recommended Reading

- Sabrin, K, Dovrolis, C. [The Hourglass Effect in Hierarchical Dependency Networks](https://www.cambridge.org/core/journals/network-science/article/hourglass-effect-in-hierarchical-dependency-networks/DDBCA83D16CA74B827DAB66A98CC906A) , <https://www.cambridge.org/core/journals/network-science/article/hourglass-effect-in-hierarchical-dependency-networks/DDBCA83D16CA74B827DAB66A98CC906A>., Journal of Network Science, (2017)
- Faskowitz, J., Yan, X., Zuo, X. *et al.* [Weighted Stochastic Block Models of the Human Connectome across the Life Span](https://www.nature.com/articles/s41598-018-31202-1) , <https://www.nature.com/articles/s41598-018-31202-1>., Sci Rep 8, 12997 (2018).

Assignments

Lesson-6: Quiz

Lesson-6: Knowledge Check